Unscented Kalman Filter (UKF) Bring-Up Guide

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1 Introduction to the Unscented Transformation

The Unscented Kalman Filter (UKF) addresses the primary weakness of the Extended Kalman Filter (EKF): the requirement for *Linearization*. While the EKF approximates non-linear functions using a first-order Taylor series expansion (Jacobians), the UKF utilizes the **Unscented Transformation** (UT).

The UT operates on the principle that it is easier to approximate a probability distribution than it is to approximate an arbitrary non-linear function. By representing the state distribution with a minimal set of carefully chosen sample points, called **Sigma Points**, the UKF captures the mean and covariance of the state to the 3rd order of the Taylor series for Gaussian inputs.

2 The Mathematical Framework

For a state vector $\mathbf{x} \in \mathbb{R}^L$ with covariance $\mathbf{P}$, the UKF generates $2L+1$ sigma points $\mathcal{X}_i$.

2.1 Sigma Point Generation

The spread of the sigma points is determined by the scaling parameter $\lambda = \alpha^2(L+\kappa) - L$:

$$\begin{aligned}
\mathcal{X}_0 &= \bar{x} \\
\mathcal{X}_i &= \bar{x} + \left(\sqrt{(L+\lambda)\mathbf{P}}\right)_i, \quad i = 1, \ldots, L \\
\mathcal{X}_i &= \bar{x} - \left(\sqrt{(L+\lambda)\mathbf{P}}\right)_{i-L}, \quad i = L+1, \ldots, 2L
\end{aligned} \tag{1}$$

2.2 Weighting Factors

Each sigma point is assigned a mean weight (W_m) and a covariance weight (W_c):

$$W_m^{(0)} = \frac{\lambda}{L+\lambda}, \quad W_c^{(0)} = \frac{\lambda}{L+\lambda} + (1 - \alpha^2 + \beta) \tag{2}$$

$$W_m^{(i)} = W_c^{(i)} = \frac{1}{2(L+\lambda)}, \quad i = 1, \ldots, 2L \tag{3}$$

3 UKF Bring-Up: Critical Success Factors

A successful UKF "Bring-Up" on embedded hardware like the `STM32F401` requires strict adherence to numerical stability. Unlike the EKF, a UKF will "explode" if the covariance matrix $\mathbf{P}$ loses its *Positive-Definiteness*. Developers must monitor the Cholesky decomposition ($\sqrt{\mathbf{P}}$) as the primary diagnostic for filter health.

4 Embedded Data Structures and Memory Layout

To implement the UKF on the `Cortex-M4` FPU, we utilize a structured approach that separates the *State Vector* from the *Numerical Covariance* descriptors. This allows for high-speed matrix operations using the `ARM CMSIS-DSP` library.

4.1 State Vector Definition ($L = 7$)

The filter tracks the 4-element unit quaternion q and the 3-axis gyroscope bias b_ω. The dimensionality $L = 7$ dictates the number of sigma points required for the Unscented Transformation:

$$N_\sigma = 2L + 1 = 15 \tag{4}$$

4.2 The `UKF_Context` Structure

The system state and weighting factors are encapsulated in a persistent context. This structure maintains both the Square-Root (S) and Standard (P) representations for comparative analysis:

- **State (`x`):** Current estimate $[q_{x,y,z,w}, b_{\omega x,y,z}]^T$.
- **Weights (`W_m`, `W_c`):** Pre-calculated UT weighting factors based on α and β.
- **Covariance Descriptors:** `arm_matrix_instance_f32` handles metadata (rows, columns, and pointer to `pData`) for the CMSIS-DSP functions.

4.3 Memory Optimization: The QR Workspace

For the SR-UKF implementation, an augmented matrix $\mathbf{A} \in \mathbb{R}^{21\times 7}$ is utilized for the Householder QR decomposition. This dimensionality represents the $2L$ sigma deviations combined with the L-dimensional process noise floor:

$$\text{QR_ROWS} = 2L + L = 21 \tag{5}$$

Storing these matrices in a flat `float32_t` array ensures sequential memory access, which is critical for cache hits on the `STM32F401`.

5 Filter Initialization and Parameter Scaling

The initialization of the UKF involves defining the initial state, the process noise floor, and the scaling parameters that govern the spread of the Unscented Transformation.

5.1 Unscented Transformation Parameters

The "spread" of the sigma points is controlled by three primary constants:

- **Alpha ($\alpha = 0.1$):** Determines the spread of sigma points around the mean. A small value (e.g., $10^{-3} \dots 1$) is used to minimize the capture of higher-order non-linearities.
- **Beta ($\beta = 2.0$):** Incorporates prior knowledge of the state distribution. For Gaussian distributions, $\beta = 2.0$ is mathematically optimal.
- **Kappa ($\kappa = 0$):** A secondary scaling parameter, typically set to 0 for state estimation.

5.2 Weighting and Denominator Guard

The weights for the mean (W_m) and covariance (W_c) are calculated based on $\lambda = \alpha^2(L + \kappa) - L$. To prevent numerical instability during division on the `STM32F401`, a *Denominator Guard* is implemented:

$$\text{denom} = L + \lambda, \quad \text{if } |\text{denom}| < \epsilon \implies \text{denom} = \epsilon \tag{6}$$

5.3 Covariance Matrix Synchronization

To maintain mathematical parity between the Square-Root (SR-UKF) and Standard UKF hypotheses, the initial covariance $\mathbf{P}$ and its Cholesky factor $\mathbf{S}$ must satisfy the identity $\mathbf{P} = \mathbf{S}\mathbf{S}^T$.

Initializing $\mathbf{S} = 0.5\mathbf{I}$ results in an initial uncertainty $\mathbf{P} = 0.25\mathbf{I}$. This "Initial Guess" represents high confidence in the identity quaternion $[0, 0, 0, 1]^T$ and the initial zero-bias estimate.

5.4 Process Noise Mapping

The process noise $\mathbf{Q}$ (Standard) and its square-root $\mathbf{Q}_s$ (SR-UKF) are synchronized to ensure identical state "growth" over time:

$$\mathbf{Q}_{ii} = (\mathbf{Q}_{s,ii})^2 \tag{7}$$

Setting $\mathbf{Q}_{s,ii} = 0.005$ yields a variance $\mathbf{Q}_{ii} = 2.5 \cdot 10^{-5}$, providing a stable "Random Walk" for the gyroscope bias states.

6 Sigma Point Generation and Gamma Scaling

The core of the Unscented Transformation is the generation of $2L + 1$ sigma points $\mathcal{X}$. These points represent the discrete distribution of the state $\mathbf{x}$ and its uncertainty $\mathbf{S}$.

6.1 The Gamma Scaling Factor

The "spread" of the points away from the mean is determined by the coefficient γ. In the Square-Root formulation, γ is derived from the square root of the augmented state dimension and the scaling parameter λ:

$$\gamma = \sqrt{L + \lambda} \tag{8}$$

By scaling the columns of the Cholesky factor $\mathbf{S}$ by γ, we generate the symmetric spread points that capture the 2nd-order moments of the distribution.

6.2 Sigma Point Mapping Logic

The 15 sigma points ($L = 7$) are stored in a flat array of 105 elements (15×7). The mapping follows a specific symmetry:

- **Point 0:** The current mean state $\bar{\mathbf{x}}$.
- **Points** $1 \ldots L$**:** The positive "offsets" $(\bar{\mathbf{x}} + \gamma \mathbf{S}_i)$.
- **Points** $L + 1 \ldots 2L$**:** The negative "offsets" $(\bar{\mathbf{x}} - \gamma \mathbf{S}_i)$.

6.3 Embedded Implementation: Column-Wise Access

The implementation utilizes the column-wise structure of the Cholesky factor $\mathbf{S}$. For each dimension i, the j-th element of the i-th sigma point is calculated as:

$$\mathcal{X}_{i,j} = \mathbf{x}_j \pm \gamma \cdot \mathbf{S}_{j,i} \tag{9}$$

This approach ensures that the "uncertainty cloud" is propagated through the non-linear dynamics in the subsequent prediction step.

7 Non-Linear Kinematics and Observation Models

The predictive power of the UKF relies on the fidelity of the transition function $f(x)$ and the observation function $h(x)$. For high-dynamic UAV flight, these are implemented via the Exponential Map and a Gravitational Reference model.

7.1 Rotational Kinematics: The Exponential Map

Traditional Euler integration ($q_{k+1} = q_k + \dot{q}dt$) "steps off" the unit sphere, leading to magnitude decay and numerical drift. To preserve geometric integrity, we utilize the **Exponential Map**, which treats rotation as a displacement along the manifold:

$$\Delta\mathbf{q} = \begin{bmatrix} \frac{\boldsymbol{\omega}}{\|\boldsymbol{\omega}\|} \sin\left(\frac{\|\boldsymbol{\omega}\|\Delta t}{2}\right) \\ \cos\left(\frac{\|\boldsymbol{\omega}\|\Delta t}{2}\right) \end{bmatrix} \tag{10}$$

The state is propagated via the Hamiltonian product: $\mathbf{x}_{out} = \mathbf{x}_{in} \otimes \Delta\mathbf{q}$. This method maintains the unit-norm constraint even during high-rate angular velocities (e.g., $500°/s$), where Euler methods typically fail.

7.2 Gravitational Observation Model

The accelerometer measurement is modeled by rotating the stationary gravity vector $[0, 0, 1]^T$ from the inertial frame to the body frame using the current quaternion estimate. The predicted measurement $\hat{z}$ is derived as:

$$\hat{\mathbf{z}} = \begin{bmatrix} 2(xz - wy) \\ 2(wx + yz) \\ w^2 - x^2 - y^2 + z^2 \end{bmatrix} \tag{11}$$

This allows the UKF to calculate the *Innovation* (the difference between predicted and actual gravity), which is subsequently used to correct the orientation and gyroscope biases.

7.3 Small-Angle Approximation Guard

To prevent numerical singularity (division by zero) during periods of zero angular velocity, the implementation includes a transcendental limit check:

$$\text{if } \|\boldsymbol{\omega}\| < \epsilon \implies \Delta\mathbf{q} = [0, 0, 0, 1]^T \tag{12}$$

This ensures that the filter remains stable and "identity-mapped" when the drone is stationary.

8 State Propagation and Covariance Reconstruction

The prediction phase of the Standard UKF involves the transformation of the sigma point cloud through the non-linear process model and the subsequent reconstruction of the state mean and covariance.

8.1 Numerical Stability: The Symmetry Guard

On 32-bit floating-point architectures like the `Cortex-M4`, rounding errors in the outer product summation can cause the covariance matrix $\mathbf{P}$ to become asymmetric or non-positive definite. To prevent filter divergence, a *Symmetry Guard* is enforced before the Cholesky decomposition:

$$\mathbf{P}_{i,j} = \mathbf{P}_{j,i} = \frac{\mathbf{P}_{i,j} + \mathbf{P}_{j,i}}{2}, \quad \mathbf{P}_{i,i} = \max(\mathbf{P}_{i,i}, \epsilon) \tag{13}$$

This ensures the matrix remains a valid covariance representation, allowing for a stable $\mathbf{S} = \text{chol}(\mathbf{P})$ update.

8.2 Weighted Mean and Quaternion Normalization

The predicted state $\hat{\mathbf{x}}$ is the weighted sum of the propagated sigma points $\mathcal{X}^*$. Unlike Euclidean states, the quaternion component must be normalized after the summation to ensure it remains on the unit sphere:

$$\hat{\mathbf{x}} = \sum_{i=0}^{2L} W_m^{(i)} f(\mathcal{X}_i, \mathbf{u}), \quad \mathbf{q}_{norm} = \frac{\hat{\mathbf{q}}}{\|\hat{\mathbf{q}}\|} \tag{14}$$

Failure to normalize individual sigma points before the summation can lead to "energy leakage," where the predicted gravity vector $\hat{z}$ loses magnitude, resulting in erroneous measurement innovations.

8.3 Covariance Reconstruction and Process Noise

The predicted covariance $\hat{\mathbf{P}}$ is reconstructed via the weighted outer product of the state deviations. The process noise $\mathbf{Q}$ is then injected into the diagonal to represent the uncertainty growth over the time step Δt:

$$\hat{\mathbf{P}} = \left(\sum_{i=0}^{2L} W_c^{(i)} (\mathcal{X}_i^* - \hat{\mathbf{x}})(\mathcal{X}_i^* - \hat{\mathbf{x}})^T \right) + \mathbf{Q} \tag{15}$$

8.4 Future Work: Dynamic Process Noise (Q)

While a static diagonal $\mathbf{Q}$ is sufficient for stable flight, future iterations may implement *Dynamic Process Noise*. By scaling $\mathbf{Q}$ based on motor RPM or vibration intensity (derived from high-pass filtered IMU data), the filter can "de-tune" its trust in the gyro integration during high-vibration events, further reducing orientation drift.

9 Measurement Update and Bayesian Correction

The update phase incorporates physical sensor data (Accelerometer) to correct the propagated state. In the Standard UKF, this is achieved by computing the

Innovation Covariance and the *Kalman Gain*.

9.1 Innovation and Cross-Covariance

The predicted measurement $\hat{\mathbf{z}}$ is the weighted mean of the measurement sigma points. We then calculate the Innovation Covariance $\mathbf{S}_{zz}$ and the Cross-Covariance $\mathbf{P}_{xz}$:

$$\mathbf{S}_{zz} = \left(\sum_{i=0}^{2L} W_c^{(i)} (\mathcal{Z}_i - \hat{\mathbf{z}})(\mathcal{Z}_i - \hat{\mathbf{z}})^T \right) + \mathbf{R} \tag{16}$$

$$\mathbf{P}_{xz} = \sum_{i=0}^{2L} W_c^{(i)} (\mathcal{X}_i - \bar{\mathbf{x}})(\mathcal{Z}_i - \hat{\mathbf{z}})^T \tag{17}$$

Where $\mathbf{R}$ represents the measurement noise floor. Setting $\mathbf{R} = 0.01$ (equivalent to a 0.1 standard deviation) provides a stable balance between gyro integration and accelerometer leveling.

9.2 Kalman Gain and State Update

The Kalman Gain $\mathbf{K}$ determines how much the filter "trusts" the sensor error (Innovation). It is solved via the matrix product:

$$\mathbf{K} = \mathbf{P}_{xz} \mathbf{S}_{zz}^{-1} \tag{18}$$

While the `arm_mat_inverse_f32` function is used for the 3×3 inversion in the primary update, the state correction is applied as:

$$\mathbf{x}_{new} = \mathbf{x}_{old} + \mathbf{K}(\mathbf{z}_{raw} - \hat{\mathbf{z}}) \tag{19}$$

9.3 Covariance Downdate and Stability Forcing

The uncertainty $\mathbf{P}$ is reduced after a successful measurement update. To prevent numerical "hanging" (where $\mathbf{P}$ remains too large, causing jitter), a *Stability Factor* is applied:

$$\mathbf{P}_{new} = (\mathbf{P}_{old} - \mathbf{K}\mathbf{S}_{zz}\mathbf{K}^T) \cdot 0.98 \tag{20}$$

This 2% artificial contraction ensures the filter converges rapidly to the identity state when the drone is stationary.

9.4 IMM Likelihood via Cramer's Rule

For the Interacting Multiple Model (IMM) selection logic, the Gaussian likelihood Λ must be calculated. To maximize performance on the `Cortex-M4`, a fast 3×3 inverse and determinant are solved manually via Cramer's Rule and Sarrus' Rule:

$$\det(\mathbf{S}_{zz}) = a(ei - fh) - b(di - fg) + c(dh - eg) \tag{21}$$

This avoids the overhead of the general-purpose `CMSIS` library calls for the probability density exponent, ensuring the IMM logic remains within the 1.6 ms execution budget.

10 Real-Time Telemetry and Hardware Profiling

To validate the UKF's performance in high-dynamic environments, a deterministic telemetry and profiling pipeline is implemented. This allows for sub-millisecond monitoring of the state vector and the "Uncertainty Cloud" (Sigma Points) via `Unity`.

10.1 Execution Profiling via DWT

On the `Cortex-M4`, the *Data Watchpoint and Trace* (DWT) unit provides a cycle-accurate timer for measuring loop execution. This is critical for ensuring the 1.6 ms IMM execution budget is maintained.

$$\Delta t = \frac{\mathrm{DWT}_{\mathrm{end}} - \mathrm{DWT}_{\mathrm{start}}}{f_{CPU}} \tag{22}$$

10.2 USB Packet Architecture (64-Byte HID)

The telemetry stream utilizes a fixed-size 64-byte packet structure to maintain compatibility with the `USB 2.0 Full Speed` HID standard. The memory map is strictly defined to ensure zero-latency parsing:

- **Index 0:** Command ID (e.g., `SR_UKF_DATA`).
- **Index 1-4:** 32-bit Microcontroller Clock (DWT Cycles).
- **Index 5-20:** State Quaternion $[q_x, q_y, q_z, q_w]$.
- **Index 21-32:** Predicted Gravity Vector $\hat{z}$.
- **Index 33:** Sigma Point Index $(0 \ldots 14)$.
- **Index 34-49:** Current Sigma Point $[q_x, q_y, q_z, q_w]$.
- **Index 63:** 8-bit CRC for packet integrity.

10.3 Sigma Point Round-Robin Strategy

Because the full state (15×7 floats) exceeds the 64-byte USB buffer, a *Round-Robin* transmission scheme is employed. Each packet contains the full mean state but only one individual sigma point.

$$T_{refresh} = N_\sigma \cdot T_{loop} = 15 \cdot 1.25\,\mathrm{ms} = 18.75\,\mathrm{ms} \tag{23}$$

This allows the `Unity` visualization to reconstruct the 3D "Sigma Cloud" every 18.75 ms, providing a visual representation of the filter's covariance without saturating the USB bandwidth.

Table 1: Design and Tuning Parameters for the Standard UKF (7D CV Model)

Category	**Parameter**	**Symbol**	**Value**
Scaling Parameters	Sigma Point Spread	α	0.1
	Distribution Correction	β	2.0
	Secondary Scaling	κ	0.0
Covariance Initialization	Initial Estimation Error	$\mathbf{P}_0$	0.25 (σ^2)
	Process Noise Variance	$\mathbf{Q}$	2.5×10^{-5} (σ^2_{proc})
	Measurement Noise	$\mathbf{R}$	0.5
Dimensions	State Vector Size	L	7
	Sigma Point Count	$2L + 1$	15

10.4 Future Work: USB High Speed (480 Mbps)

The upcoming transition to the **STM32N6** will remove this bandwidth constraint. By utilizing *USB High Speed*, the system will support a 256-byte packet payload, enabling the transmission of all 15 sigma points in a single frame. This will eliminate the "Sigma Snail" lag and provide instantaneous visualization of the covariance manifold at 800 Hz.